Answer any **four** questions

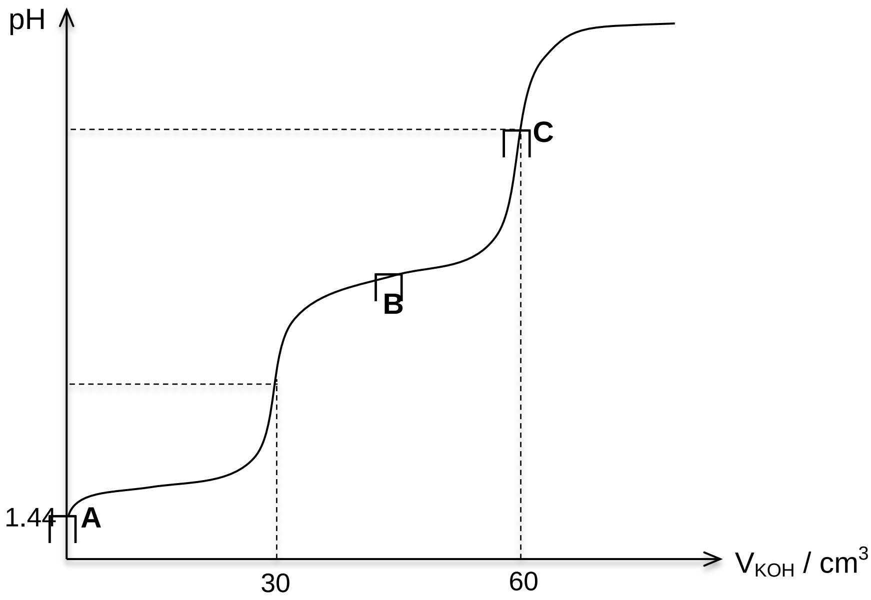
1 This question is about reactions involving the use of acids that contains sulfur.

(a) Sulfur dioxide dissolves in water to form a weak dibasic acid, H_2SO_3, which dissociates as follows:

$$H_2SO_3(aq) \rightleftharpoons H^+(aq) + HSO_3^-(aq), \quad \boldsymbol{K_{a1}} = x \text{ mol dm}^{-3}$$

$$HSO_3^-(aq) \rightleftharpoons H^+(aq) + SO_3^{2-}(aq), \quad \boldsymbol{K_{a2}} = 6.50 \times 10^{-8} \text{ mol dm}^{-3}$$

30.0 cm^3 of a 0.20 mol dm^{-3} solution of H_2SO_3 was titrated against a solution of 0.10 mol dm^{-3} potassium hydroxide. The resulting pH curve was plotted as shown below:



(i) Explain the meaning of the term *weak Bronsted acid*.

(ii) Using the pH at point **A**, calculate the value of x, the first acid dissociation constant, $\boldsymbol{K_{a1}}$, of H_2SO_3.

(iii) By writing two equations, show how the solution at point **B** acts as a buffer solution, when a small amount of acid or base is added to it.

(iv) Explain with the help of an equation why the pH at point **C** is not equal to 7.

[7]

(b) Sulfur trioxide can be absorbed in sulfuric acid to form oleum, $H_2S_2O_7$. The structure of oleum is shown below.

$$HO{-}SO_2{-}O{-}SO_2{-}OH$$

In terms of bonding and structure, explain why oleum has a higher melting point than sulfuric acid.

[3]

(c) Thiol is an organosulfur compound which is a weak monobasic acid. An example is methylthiol, CH_3SH.

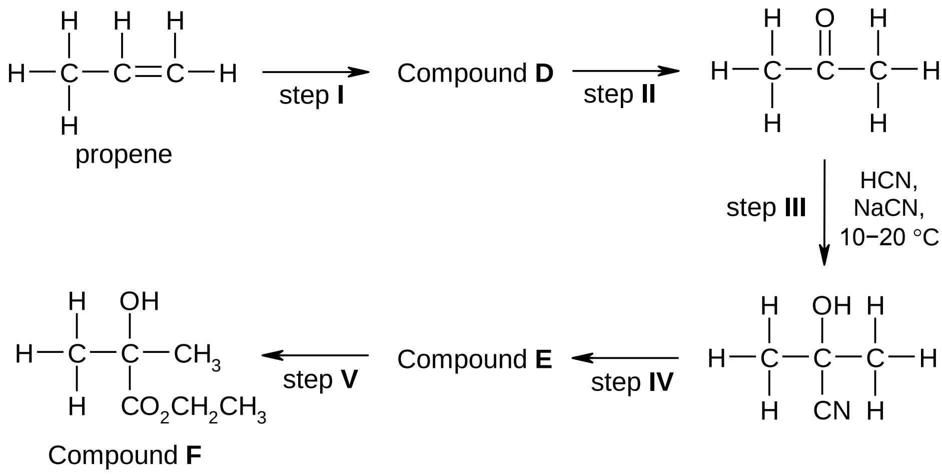
(i) What do you understand by the term *standard enthalpy change of neutralisation*?

(ii) When 50 cm^3 of 1.00 mol dm^{-3} of CH_3SH was added to 100 cm^3 of 6.00 mol dm^{-3} aqueous NaOH, a temperature rise of 2.0 °C was obtained. Assuming that there is negligible heat loss to the surroundings, what is the enthalpy change of neutralisation for this reaction?

(Density of solution may be taken as 1.00 g cm^{-3} and specific heat capacity of the solution as 4.20 J $g^{-1}K^{-1}$)

[3]

(d) Sulfuric acid is a commonly used reagent for organic synthesis. The five-step synthesis of compound **F** with propene as the starting material could involve the use of sulfuric acid in some of the steps.



(i) Draw the structural formula of compounds **D** and **E**.

(ii) Steps **I**, **II**, **IV** and **V** in the above reaction scheme all involve the use of sulfuric acid. State the reagents and conditions for each of these steps.

(iii) Name and outline the mechanism for step **III**.

[7]

[Total: 20 marks]

2 This question deals with the chemistry of the first row transition elements.

(a) Describe the colour changes observed when dilute aqueous ammonia is added to a solution containing Cu^{2+}(aq) until the ammonia is in an excess, giving the formulae of all relevant copper compounds.

[4]

(b) When black copper(II) oxide is stirred with liquid ammonia, it dissolves to give a coloured solution. During this reaction, the oxide ion is acting as a Bronsted-Lowry base.
Suggest an equation for this reaction, and suggest the colour of the solution.

[2]

(c) When concentrated hydrochloric acid is added to a solution containing Cu^{2+}(aq), the colour changes to yellow. No gas is evolved.
No such colour change occurs when concentrated sulfuric acid is added to Cu^{2+}(aq).
Dilution of the yellow solution with water produces the original pale blue colour.
Suggest an explanation of these observations.

[3]

(d) The yellow solution, sodium tetrahydroxochromate(III), $Na[Cr(OH)_4]$, can be prepared by the addition of excess sodium hydroxide to a green solution of aqueous chromium(III) chloride, $CrCl_3$.

Draw the structure of the complex in the yellow solution to illustrate the bond angle present, and suggest the shape of the complex.

[2]

(e) This question is about compound **K**, C_8H_7ON.

Compound **K** gives only **L** on heating with dilute HC*l*. **L** reacts with LiA*l*H_4 to give **M**. When **M** is passed over hot Al_2O_3, **N** is obtained. **O** is formed together with a colourless gas when **N** is heated with acidified $KMnO_4$.

O can also be prepared by treating 2-nitrobenzoic acid with Sn in concentrated HC*l*.

Deduce the structures of compounds **K**, **L**, **M**, **N** and **O**, and explain the chemistry of the reactions described.

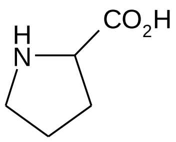
[9]

[Total: 20 marks]

3 The chemistry of iron has been extensively studied and its applications span from industrial processes to biological functions.

(a) Many proteins that have crucial roles in cellular physiology require iron to function. One example is haemoglobin, a globular protein responsible for the transport of oxygen in the blood stream.

Some of the amino acids found in haemoglobin are listed below.

valine (val)	$(CH_3)_2CHCH(NH_2)CO_2H$
glutamic acid (asp)	$HO_2CCH_2CH_2CH(NH_2)CO_2H$
serine (ser)	$HOCH_2CH(NH_2)CO_2H$
cysteine (cys)	$HSCH_2CH(NH_2)CO_2H$
proline (pro)	

(i) The amino acids whose side chains are on the outside of haemoglobin are likely to interact strongly with water molecules. Suggest which of the above amino acids are most likely to do so.

(ii) Draw the displayed formula of a section of the protein chain of haemoglobin with the sequence **ser-pro-glu** obtained in a solution at pH 13.

(iii) Most of the amino acids present in haemoglobin form α-helices.

With the aid of a diagram, describe how a polypeptide chain is held in the shape of an α-helix.

[5]

(b) The function of a protein is affected when the *quaternary structure* changes.

(i) With reference to haemoglobin, explain what is meant by the term *quaternary structure* of a protein.

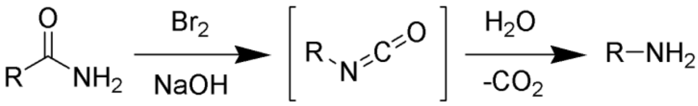
(ii) Under deoxygenated conditions, there is a small non-polar patch on the surface of the haemoglobin molecule.

In patients suffering from sickle-cell anaemia, a mutation in the genes results in glutamic acid being replaced by valine. The haemoglobin aggregates and changes the shape of the red blood cells into an abnormal sickle shape.

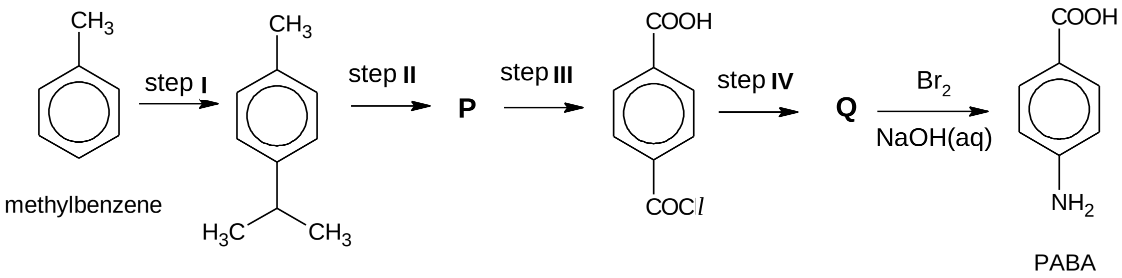
By considering the type of interactions involved, suggest how the difference in structure of the two amino acids might affect the quaternary structure of haemoglobin, hence leading to aggregation.

[4]

(c) In 1989, a group of Caltech researchers reported that iron powder can be used in the biodegradation of an azo dye derived from para-aminobenzoic acid (PABA). The industrial preparation of PABA, a type of γ-amino acid, involves Hoffmann degradation, which converts a primary amide to a primary amine.



A student proposed the following synthetic route to prepare PABA from methylbenzene.



(i) Draw the structures of **P** and **Q**.

(ii) Suggest the reagents and conditions for steps **I** and **IV**.

(iii) Suggest a simple chemical test to distinguish methylbenzene and PABA, stating the observations clearly.

[6]

(d) The following experimental steps are used to determine the composition of the elements, iron and tin, in a food can.

Step 1: 2.60 g of a piece of the food can was allowed to react with excess dilute hydrochloric acid until no more gas is evolved.

Step 2: Excess aqueous sodium hydroxide was added and a green precipitate **R** was formed which was then filtered, washed with distilled water and dried.

Step 3: The precipitate **R** was heated strongly in air to a constant mass, resulting in a brown precipitate **S**. The mass of precipitate **S** was found to be 3.20 g.

(i) With the aid of equations, explain the relevant observation in **step 1**.

(ii) State the function of NaOH in **step 2**.

(iii) Suggest an equation for **step 3**.

(iv) Hence, determine the percentage by mass of tin in the food can.

[5]

[Total: 20 marks]

4 **(a)** Magnesium nitrate and barium nitrate decompose on heating at different temperatures.

By quoting appropriate data from the *Data Booklet*, deduce which of the two nitrates decomposes at a lower temperature. Explain your answer.

[3]

(b) Dinitrogen oxide, N_2O, is used as an anaesthetic. It is manufactured by the thermal decomposition of ammonium nitrate.

$$NH_4NO_3(s) \rightarrow N_2O(g) + 2H_2O(g)$$

(i) N_2O is a linear molecule with the same number of electrons as CO_2. Draw the dot-and-cross diagram showing only the outer electrons of N_2O.

(ii) The following table lists some $\Delta H_f^\ominus$ values.

compound	$\Delta H_f^\ominus$ / kJ mol^{-1}
$NH_4NO_3(s)$	−366
$N_2O(g)$	+82
$H_2O(g)$	−242

Use the data above to calculate the standard enthalpy change for the thermal decomposition of NH_4NO_3.

(iii) By considering the entropy change and enthalpy changes, suggest why N_2O cannot be synthesised directly from its elements.

[4]

(c) At 1200 K, in the presence of a gold wire, N_2O decomposes into its elements.

The rate of decomposition of pure N_2O can be followed by measuring the total pressure, *p*, as it changes with time, *t*. In such an experiment, the total pressure increased as follows.

total pressure, *p* / kPa	25.0	27.5	30.0	32.5	34.0	35.0
time, *t* / s	0	1030	2360	4230	5870	7420
partial pressure of N_2O / kPa	25.0					

(i) Show that the partial pressure of N_2O at any time *t* is equal to (75.0 – 2*p*) kPa.

(ii) Hence calculate the partial pressures of N_2O after 1030 s, 2360 s, 4230 s, 5870 s and 7420 s.

(iii) By using a graphical method, determine the order of reaction with respect to N_2O.

(iv) Calculate the rate constant for the reaction, giving its units.

[8]

(d) 1,4-diaminobutane is also known as *putrescine* on account of its horrible smell. The compound is present in rotting flesh and also in 'bad breath'.

(i) The odour of *putrescine* can be removed by reacting it with an acid. Draw the structure of the organic product of this reaction and suggest why it has no odour?

(ii) Suggest how 1,4-diaminobutane can be synthesised from ethene. Indicate the reagents and conditions you would use for each step as well as the structures of the intermediate products. Balanced equations are not required.

[5]

[Total: 20 marks]

5 (a) The use of the *Data Booklet* is relevant to this question.

Each of the four pairs of equations below represents one reaction which does occur in accordance with the equation given and one which does not.

For **each** pair of equations, state which one represents an authentic reaction and state the type of reaction. Suggest a reason why the other reaction does not occur in practice in a similar way.

(i) $Cl_2(aq) + H_2O_2(aq) \rightarrow 2Cl^-(aq) + 2H^+(aq) + O_2(g)$

$I_2(aq) + H_2O_2(aq) \rightarrow 2I^-(aq) + 2H^+(aq) + O_2(g)$

(ii) $C_6H_5Cl(l) + OH^-(aq) \rightarrow C_6H_5OH(l) + Cl^-(aq)$

$C_6H_{11}Cl(l) + OH^-(aq) \rightarrow C_6H_{11}OH(l) + Cl^-(aq)$

(iii) $CH_4(g) + Cl_2(g) \rightarrow CH_3Cl(g) + HCl(g)$

$CH_4(g) + I_2(g) \rightarrow CH_3I(g) + HI(g)$

(iv) $CCl_4(l) + 2H_2O(l) \rightarrow CO_2(g) + 4HCl(g)$

$SiCl_4(l) + 2H_2O(l) \rightarrow SiO_2(s) + 4HCl(g)$

[8]

(b) Chlorine dioxide, ClO_2, is commonly used as a disinfectant and bleaching agent. It is a strong oxidising agent in acidic solution.

When chlorine dioxide, ClO_2, is added to a solution of Fe^{2+}, the oxidation state of chlorine decreases by 5 units and the $E^\ominus_{cell}$ of the reaction is +0.73 V.

(i) Construct a balanced half-equation for the reduction of ClO_2.

(ii) Hence construct the equation for the overall equilibrium, showing your working.

(iii) By using suitable data from the *Data Booklet*, calculate the value for the standard reduction potential of ClO_2.

(iv) Equilibrium constant, K_c, and $E^\ominus_{cell}$ are related by the following equation.

$$\ln K_c = \frac{zFE^\theta_{cell}}{RT}$$

where z is the number of moles of electrons transferred during the redox reaction and F is the Faraday's constant.

Use the equation you have written in **(b)(ii)** to decide on a suitable value for z, and hence calculate the value of K_c for the reaction between ClO_2 and Fe^{2+} at 25 ^oC. Include units for your answer.

(v) Suggest what is the significance of the magnitude of your answer in **(b)(iv)**.

[6]

(c) When solutions of sodium chloride of different concentration is electrolysed separately, different gases are produced at the anode. However, the same gas is produced at the cathode.

(i) Write a balanced ion-electron equation for the reaction taking place at the cathode when a concentrated solution of sodium chloride is electrolysed using inert electrode.

(ii) Hence, calculate the pH of the solution after a current of 0.30 A is passed through 450 cm^3 of a 3.0 mol dm^{-3} solution of sodium chloride for 20 minutes using inert electrodes.

[4]

(d) State under what conditions of temperature and pressure is a real gas such as chlorine most likely to behave like an ideal gas? Give reasons for your answers.

[2]

[Total: 20 marks]

~ END OF PAPER ~